



25801828

QP CODE: 25801828

Reg No :

Name :

M.C.A DEGREE EXAMINATION, APRIL 2025

Second Semester

MASTER OF COMPUTER APPLICATION

CORE - MCACT204 - DATA SCIENCE & BIG DATA ANALYSIS

2020 Admission Onwards

32843A35

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. What is Data Mining?
2. What are the major issues of data mining?
3. What is Data objects?
4. Define Frequent itemset and closed itemset.
5. Illustrate how association rules are generated from frequent item set of a transactional database.
6. Compare Classification and Prediction.
7. Explain credit risk management.
8. Briefly explain Big Data in healthcare.
9. What are the four V's of Big Data?
10. What are the techniques to optimize MapReduce jobs?
11. What are the functions of mapper and reducer?
12. What is the role of Hive?

(10×3=30 marks)





Part B

Answer **all** questions

Each question carries **9** marks

13. a) What are the major issues of Data Mining?

OR

b) How do we handle data cleaning process in Data Mining?

14. a) Discuss Market Basket Analysis. Briefly explain how Association rules are generated.

OR

b) Differentiate Clustering and Classification with suitable examples.

15. a) Explain Web Analytics and Credit risk management.

OR

b) Explain the different facets of data in Data Science.

16. a) Compare the distributed and parallel computing of big data with sufficient explanation.

OR

b) What is Hadoop ecosystem? Explain with the data processing elements YARN and MapReduce in detail.

17. a) Write notes on a) MapReduce framework b) Polyglot persistence

OR

b) Explain a) architecture of Hive b) Any 5 built-in functions in Hive

(5×9=45 marks)

